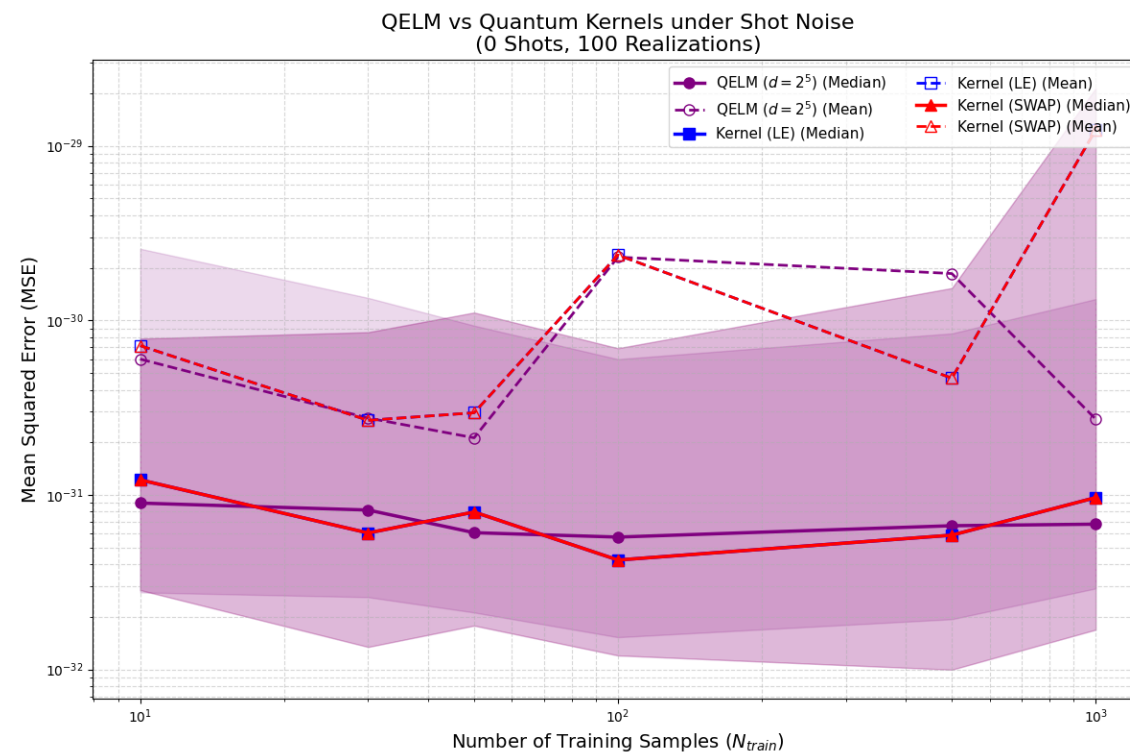
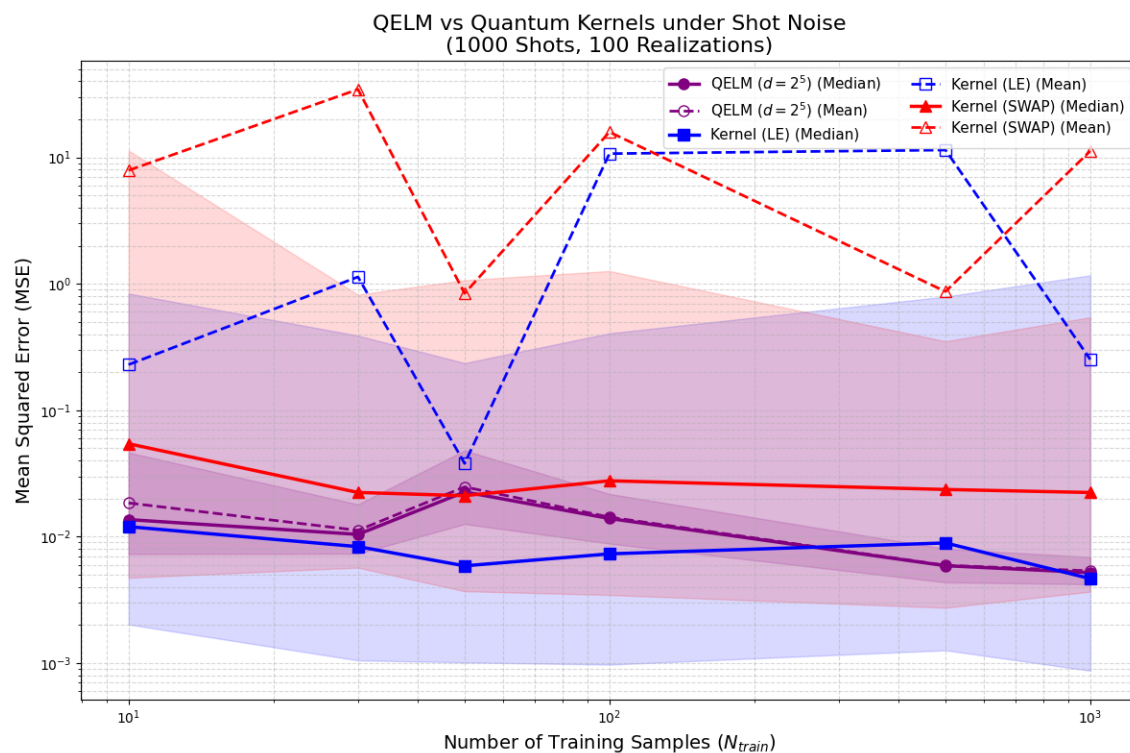


Kernel VS QELM

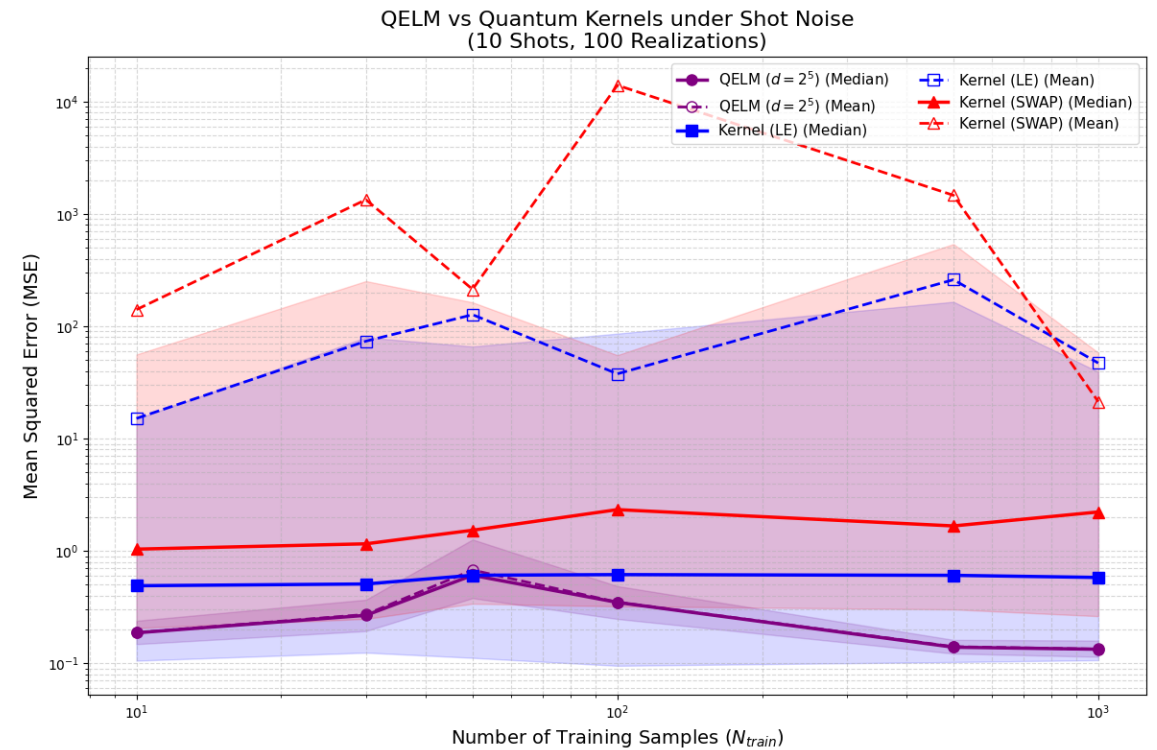
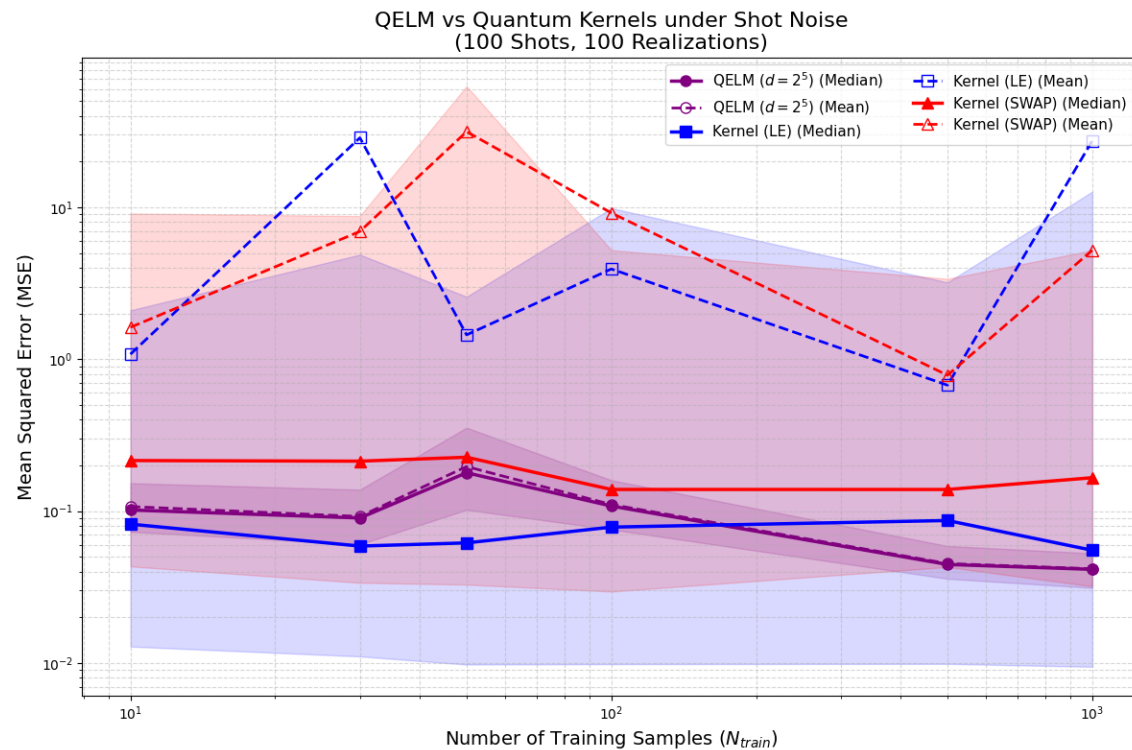
Quantum Kernel VS QELM

input qubit, reconstruct $\langle X \rangle$
, realizations = 100, rcond = default
(10^{-15}), $\lambda = 0$

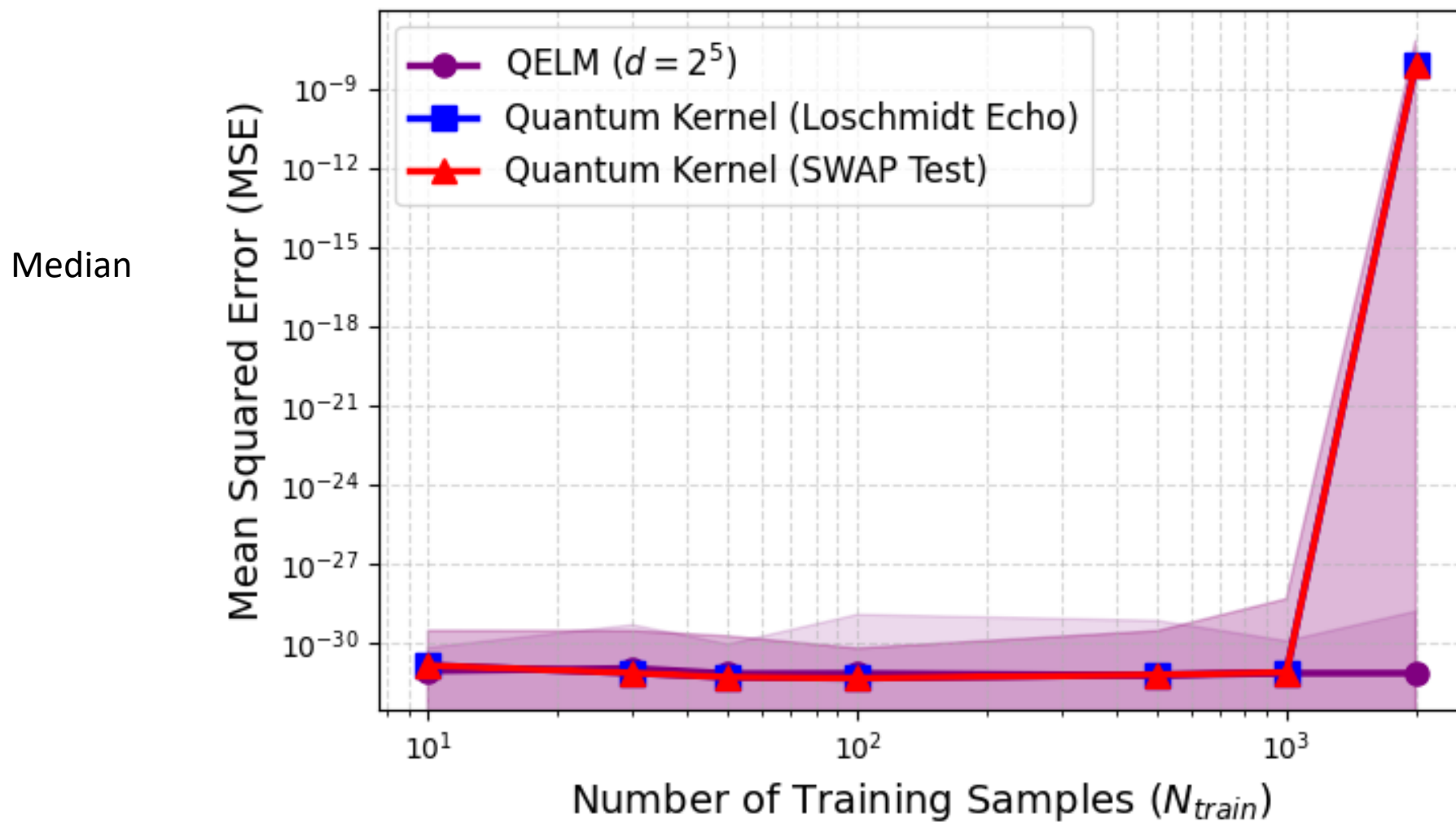


Quantum Kernel VS QELM

input qubit, reconstruct $\langle X \rangle$
, realizations = 100, rcond = default
(10^{-15}), $\lambda = 0$



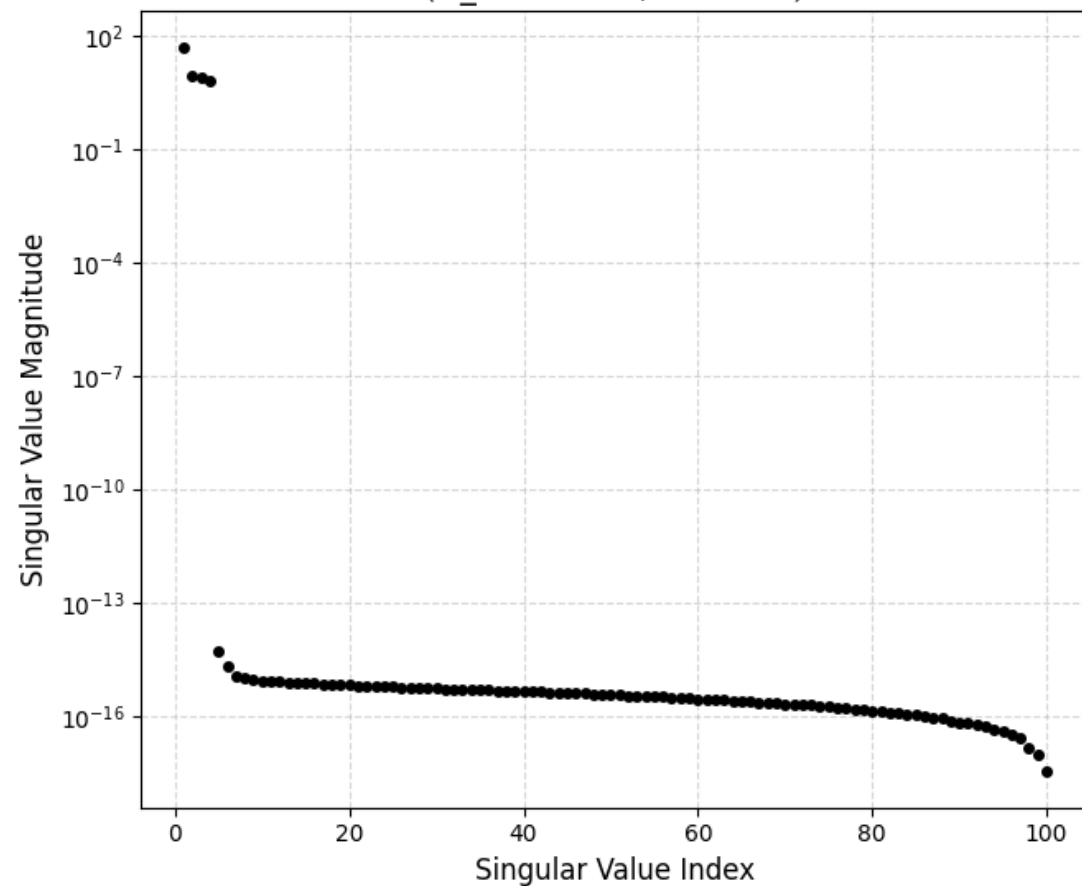
QELM vs Quantum Kernels under Shot Noise
(0 Shots, 100 Realizations)



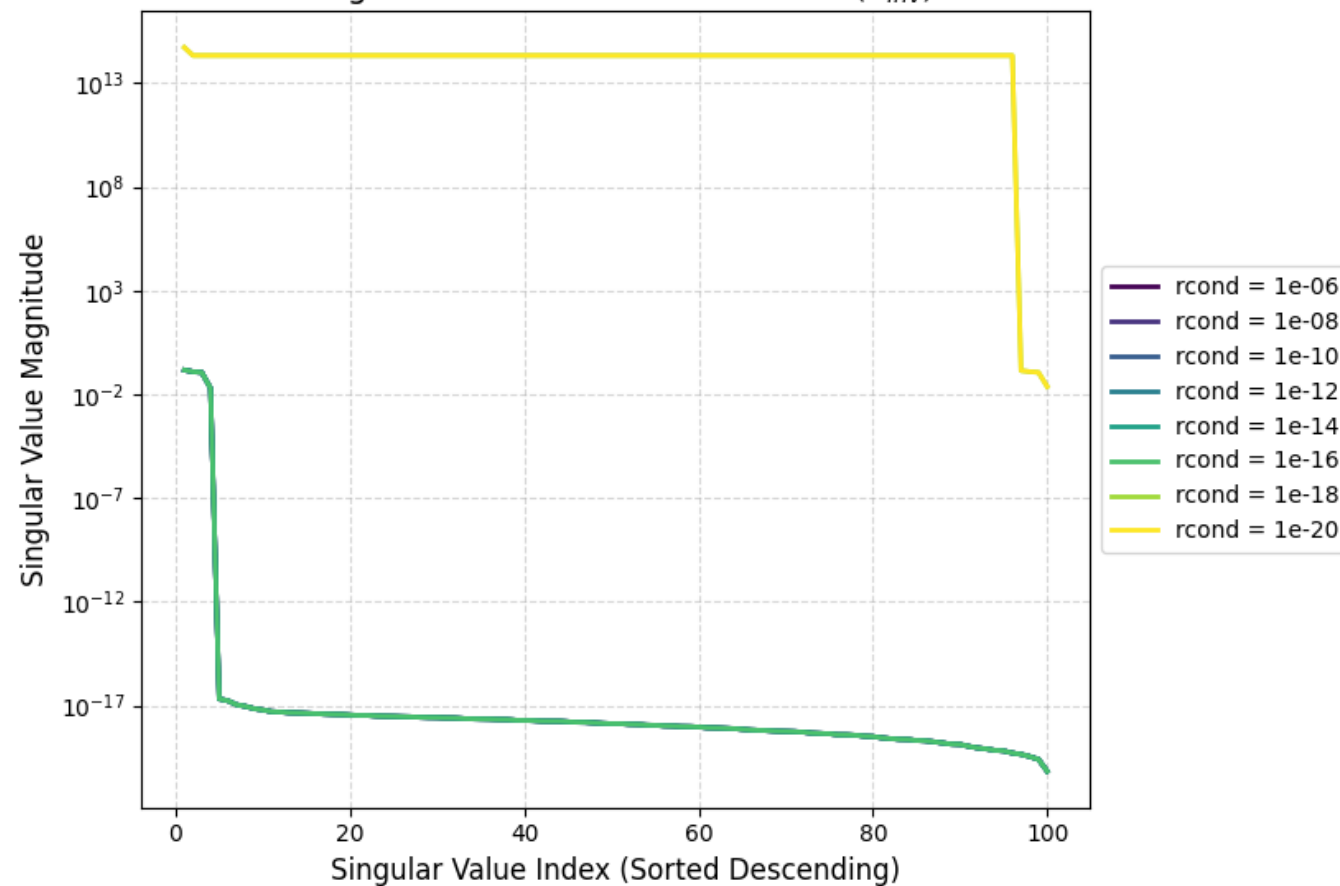
Kernel numerical instability

KernelSVD, N_train=100

Singular Values of Original Kernel Matrix K
(N_train=100, shots=0)

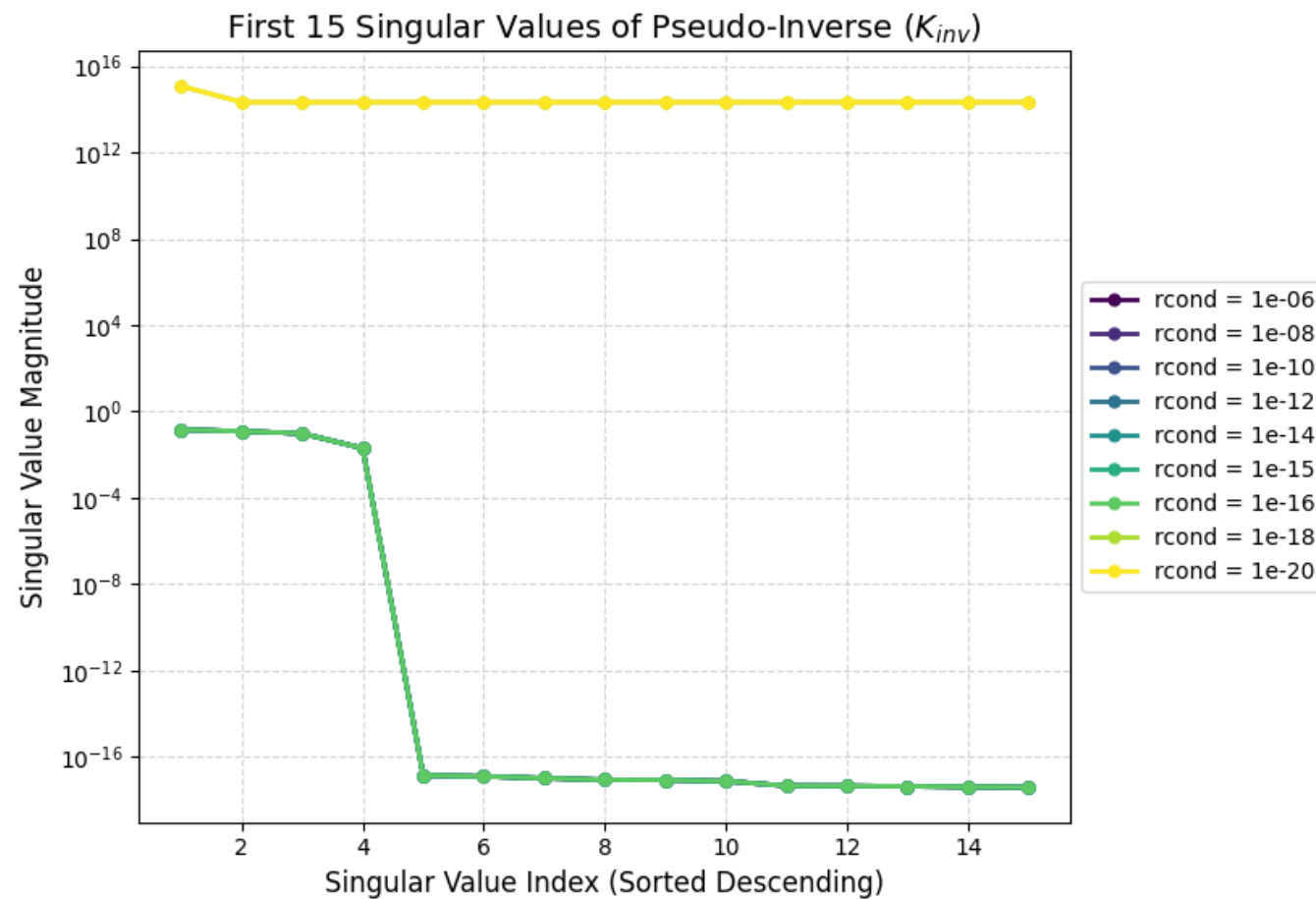
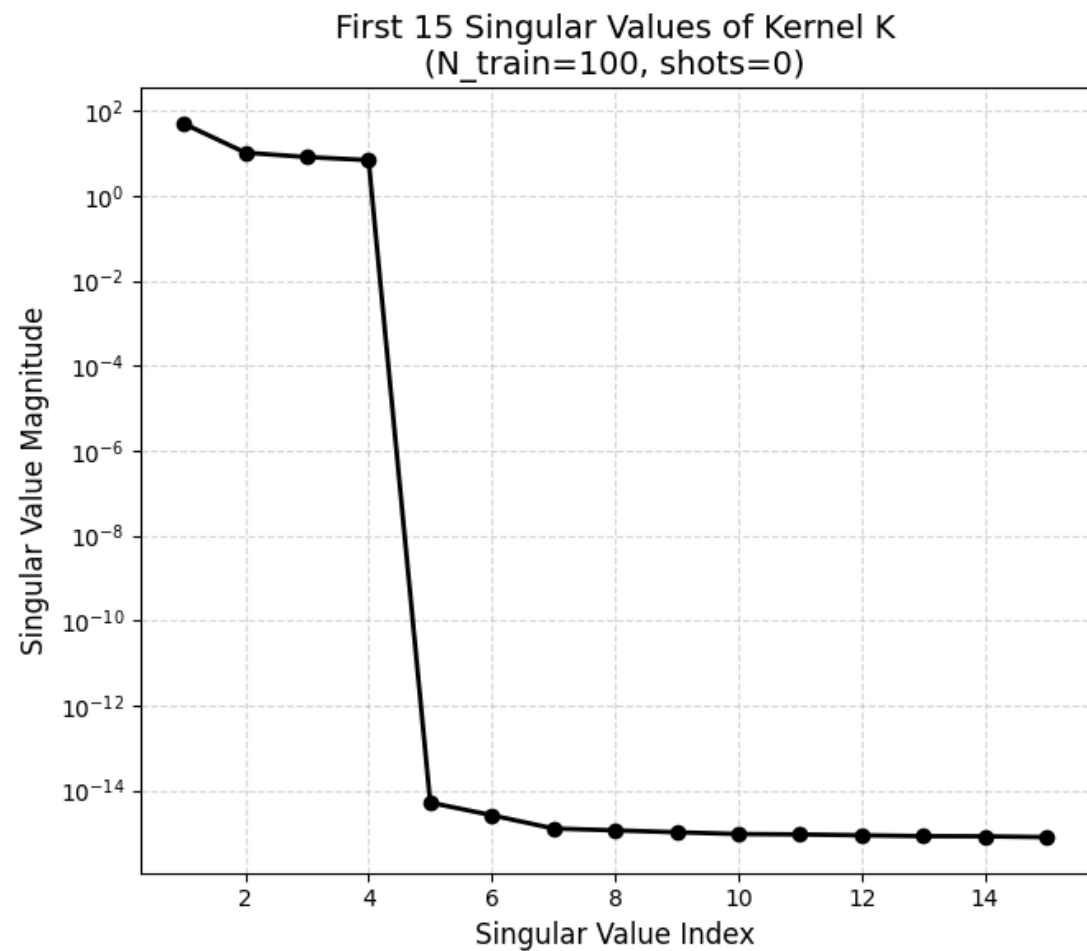


Singular Values of Pseudo-Inverse (K_{inv})



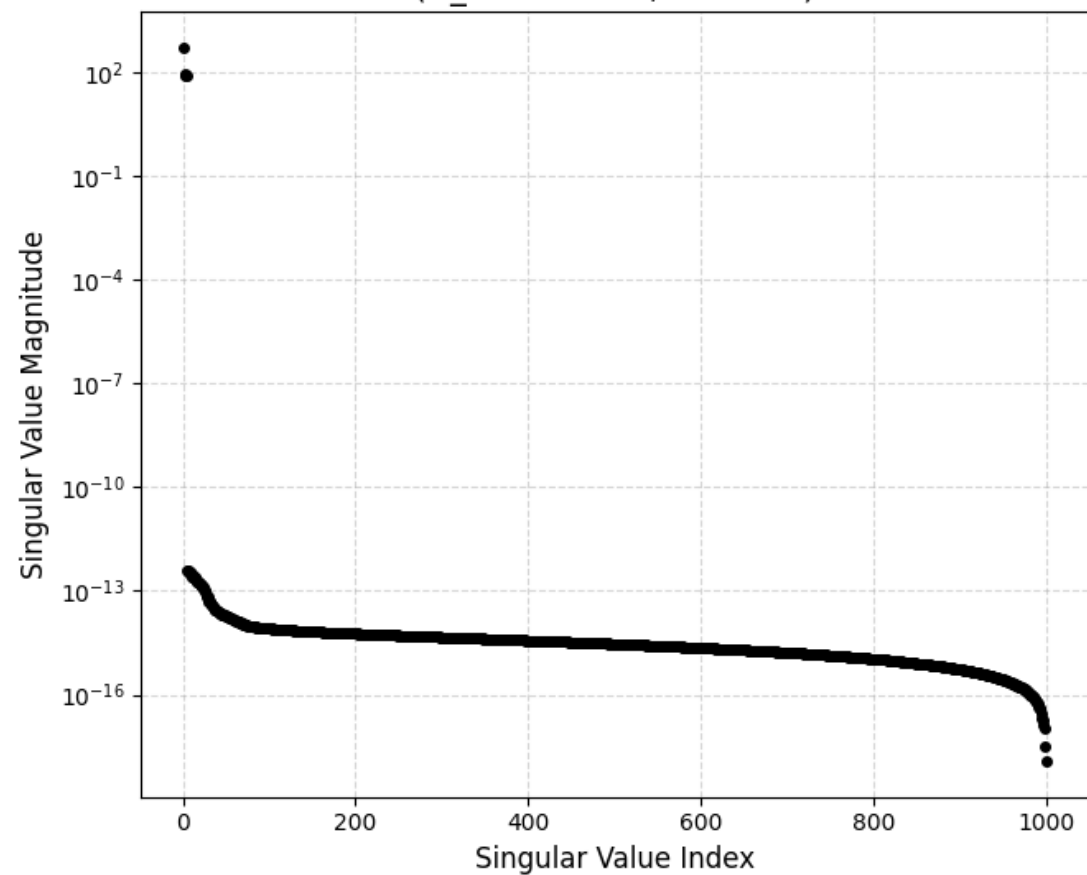
KernelSVD, N_train=100

Pinv -> rcond = 1e-15 (default)

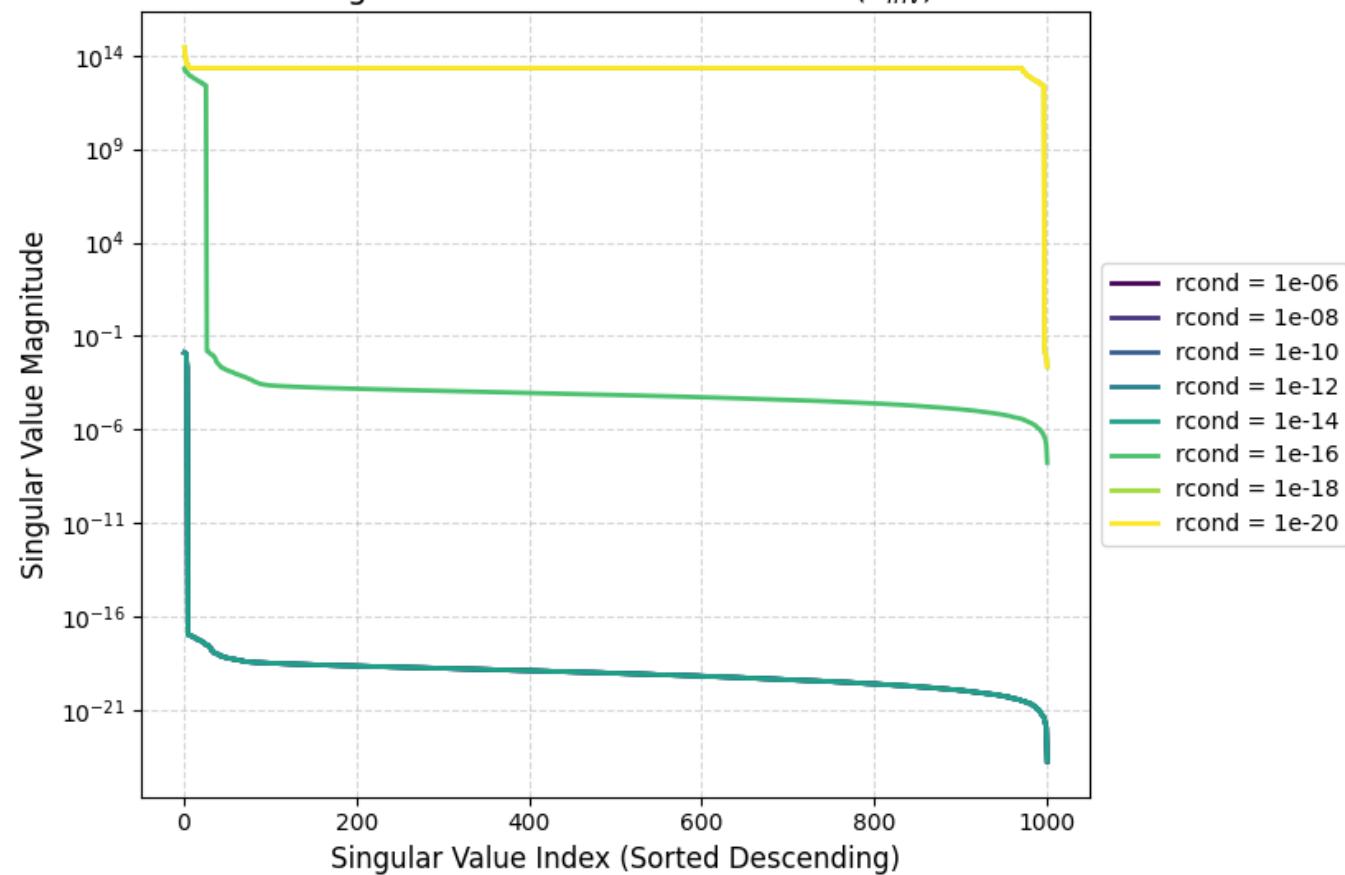


KernelSVD, N_train=1000

Singular Values of Original Kernel Matrix K
(N_train=1000, shots=0)



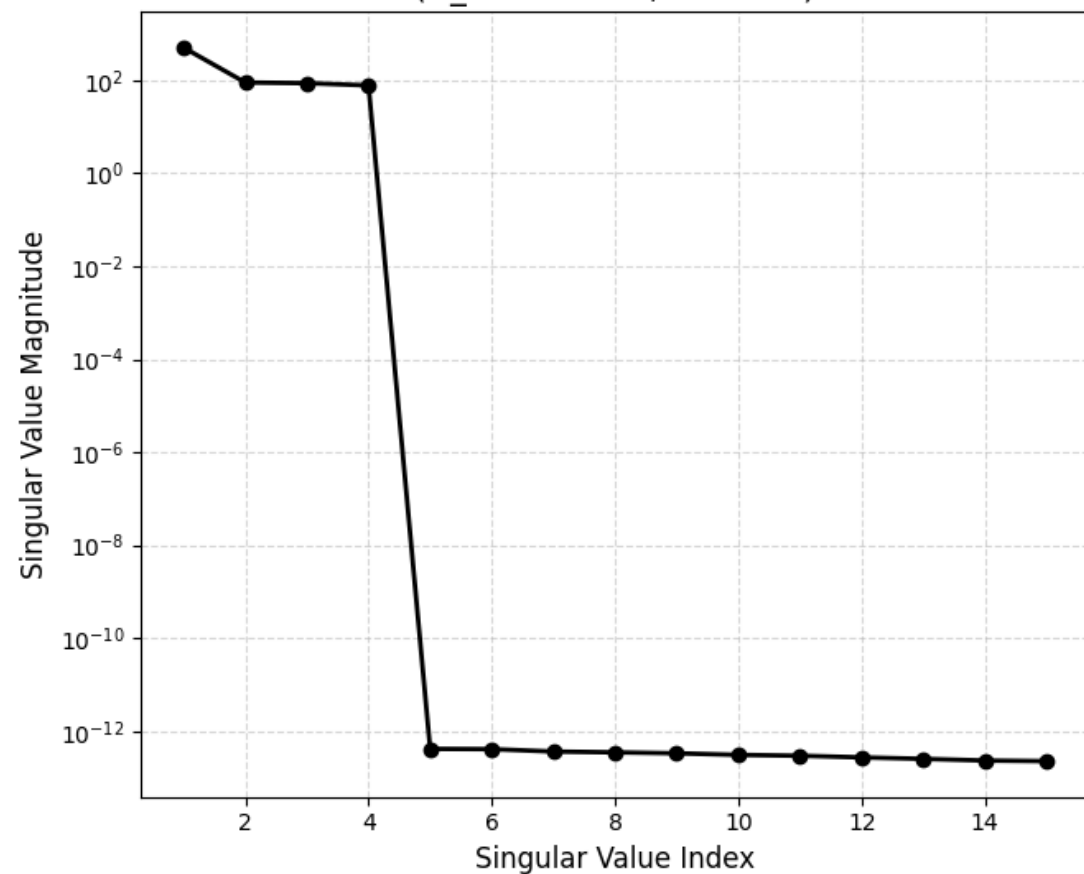
Singular Values of Pseudo-Inverse (K_{inv})



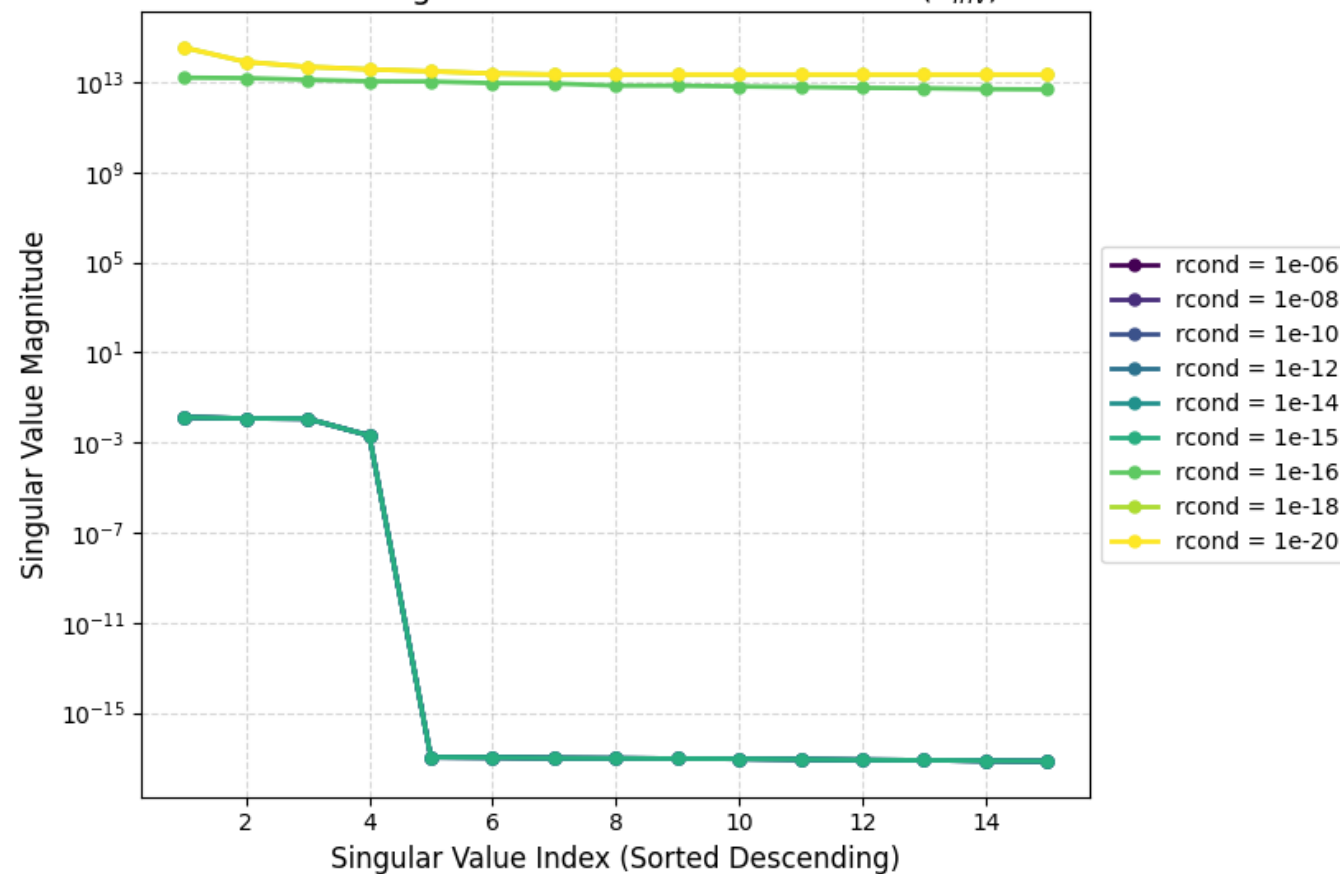
KernelSVD, N_train=1000

Pinv -> rcond = 1e-15 (default)

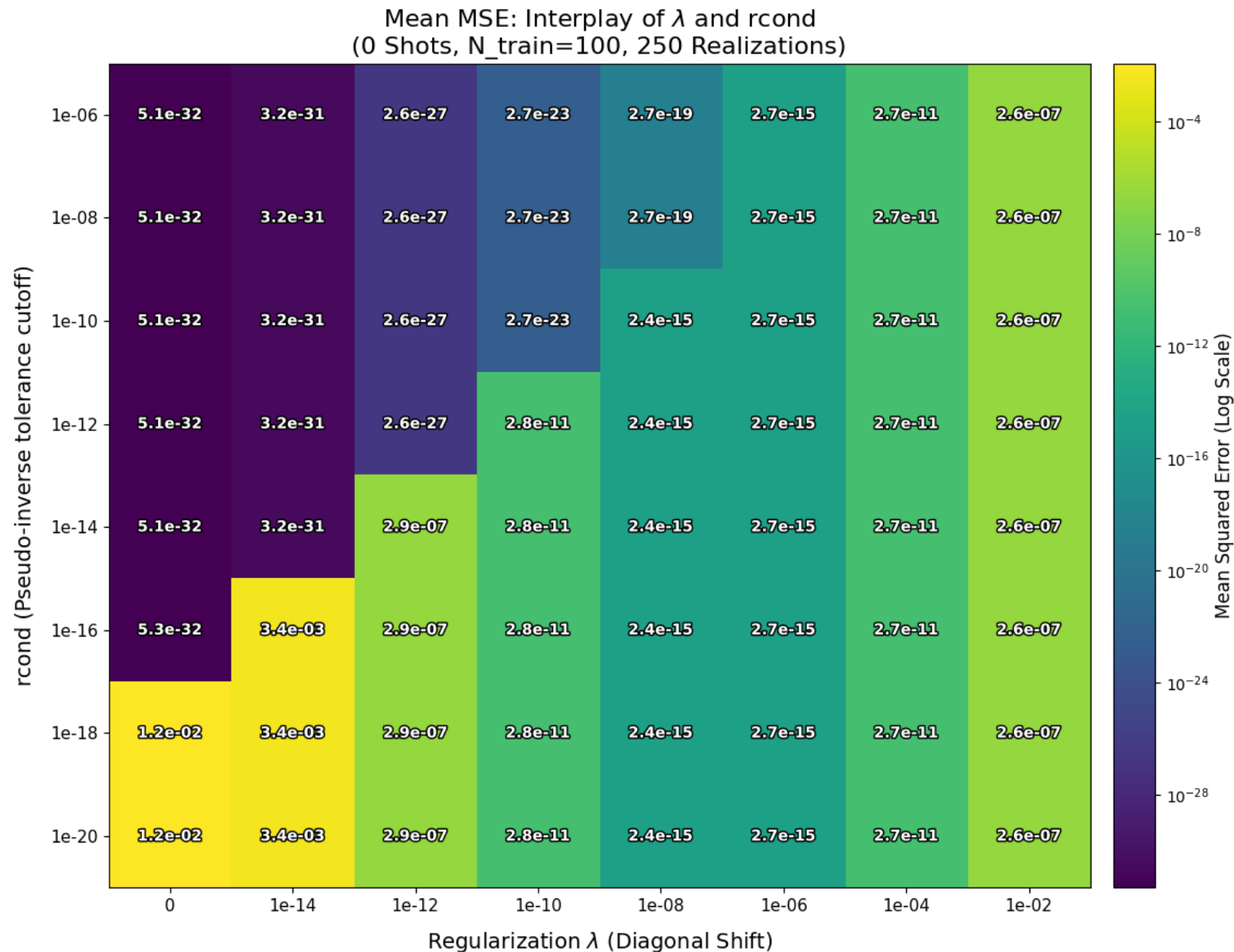
First 15 Singular Values of Kernel K
(N_train=1000, shots=0)



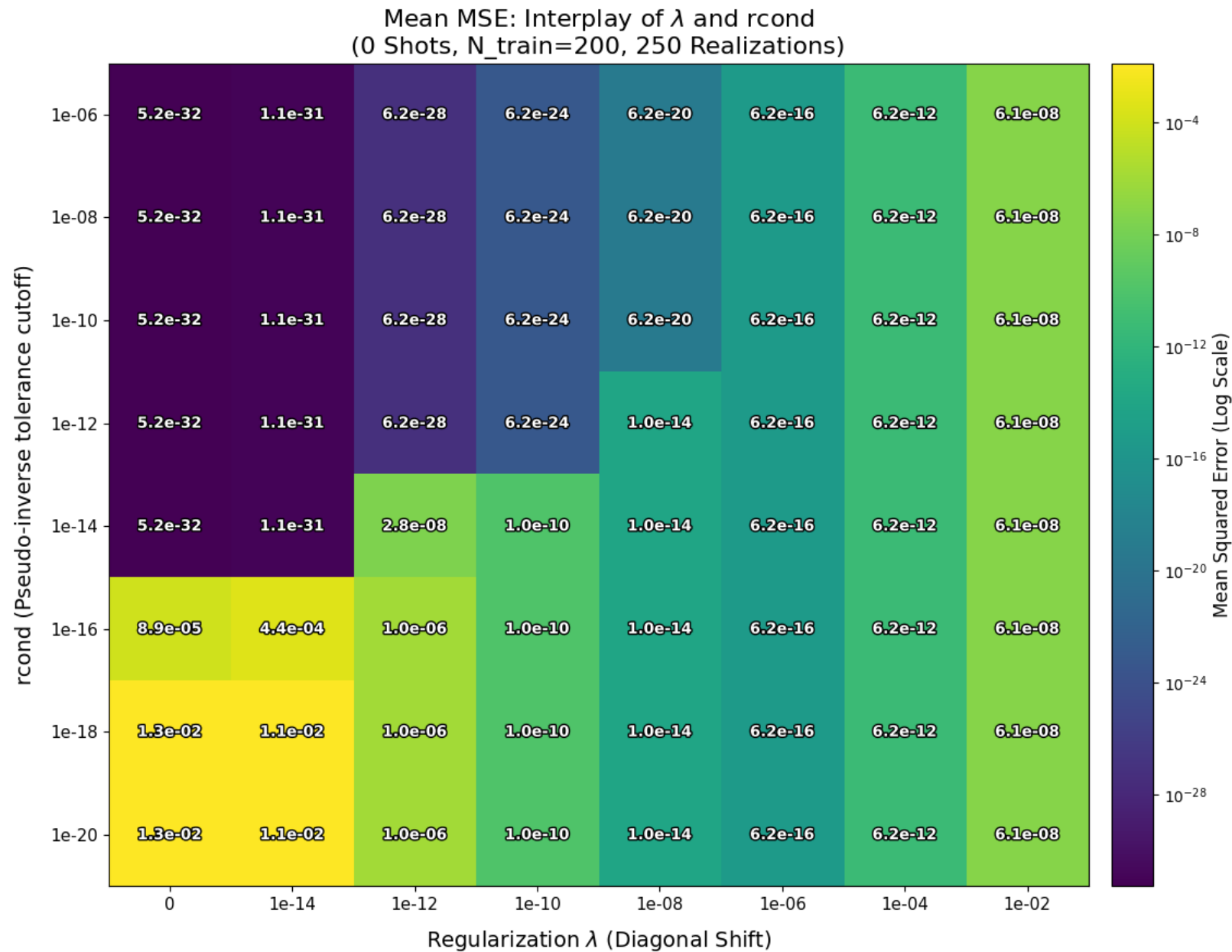
First 15 Singular Values of Pseudo-Inverse (K_{inv})



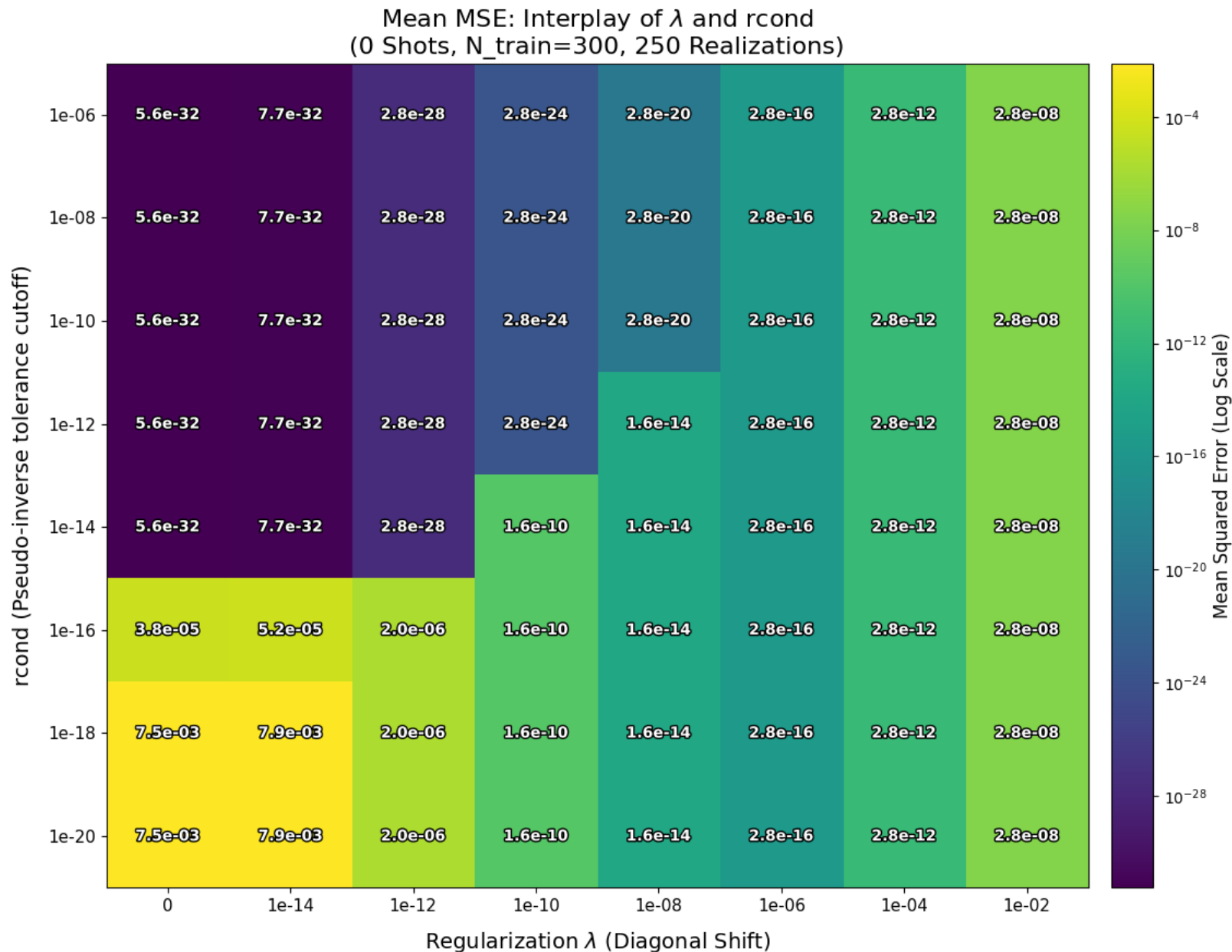
Kernel MSE, λ – rcond



Kernel MSE, λ – rcond

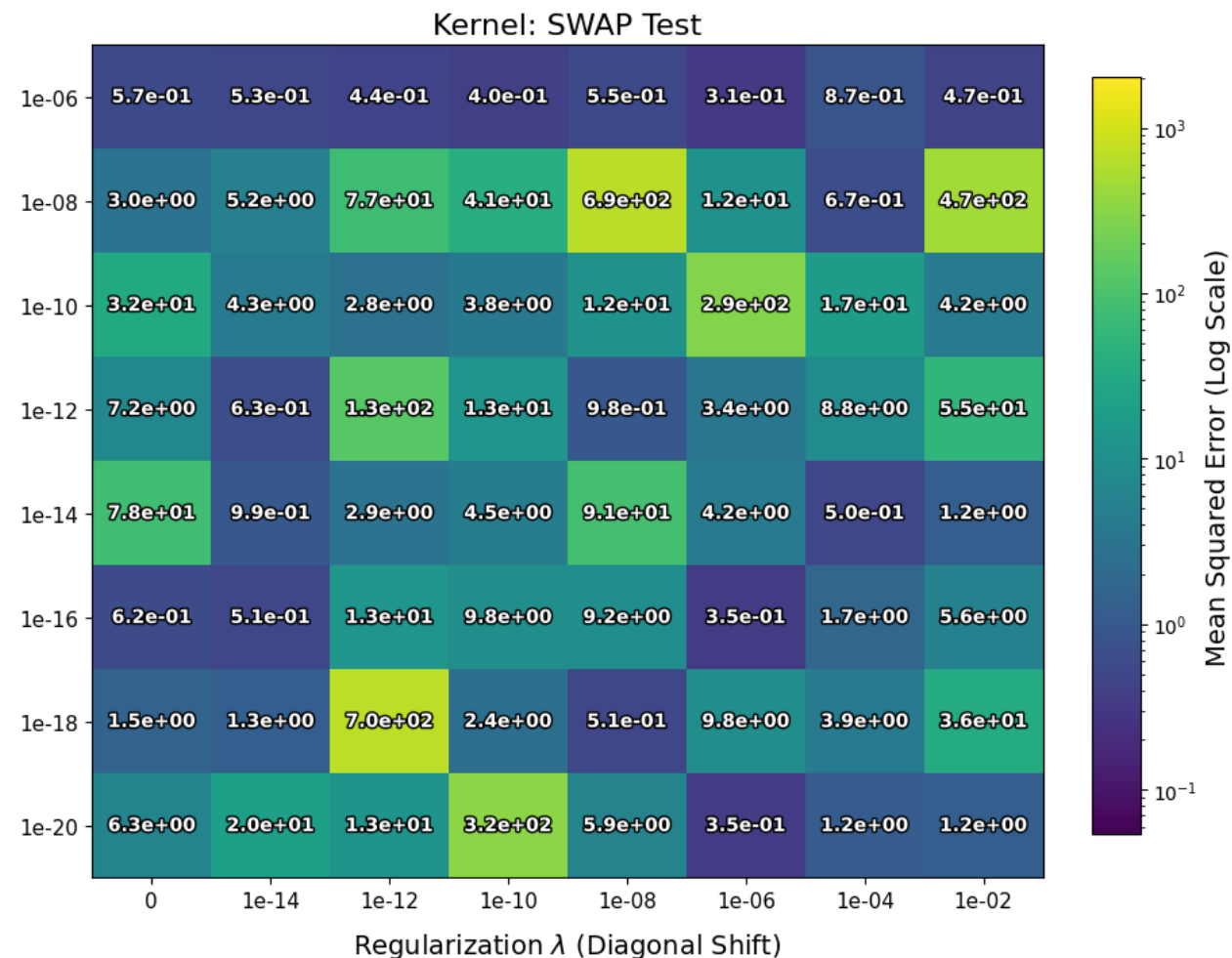
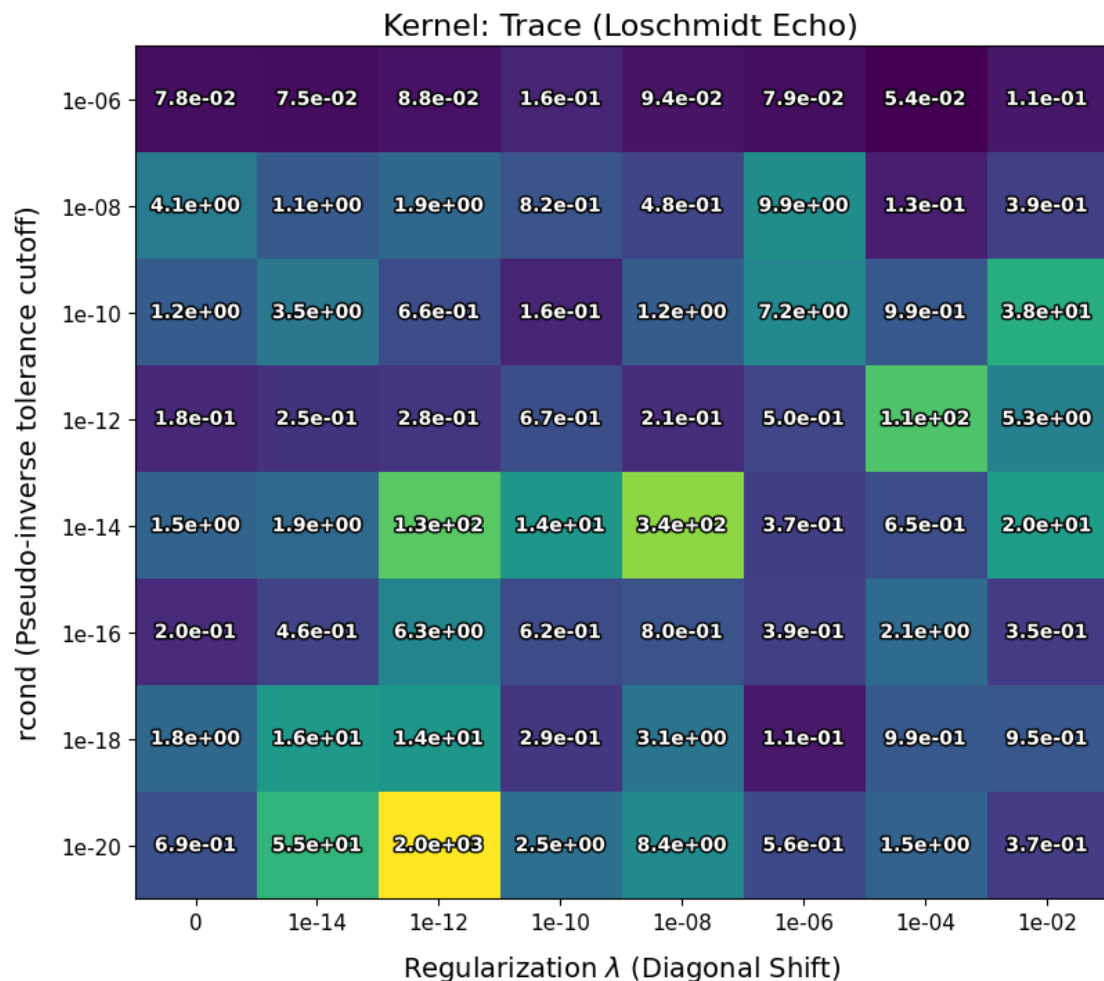


Kernel MSE, λ – rcond



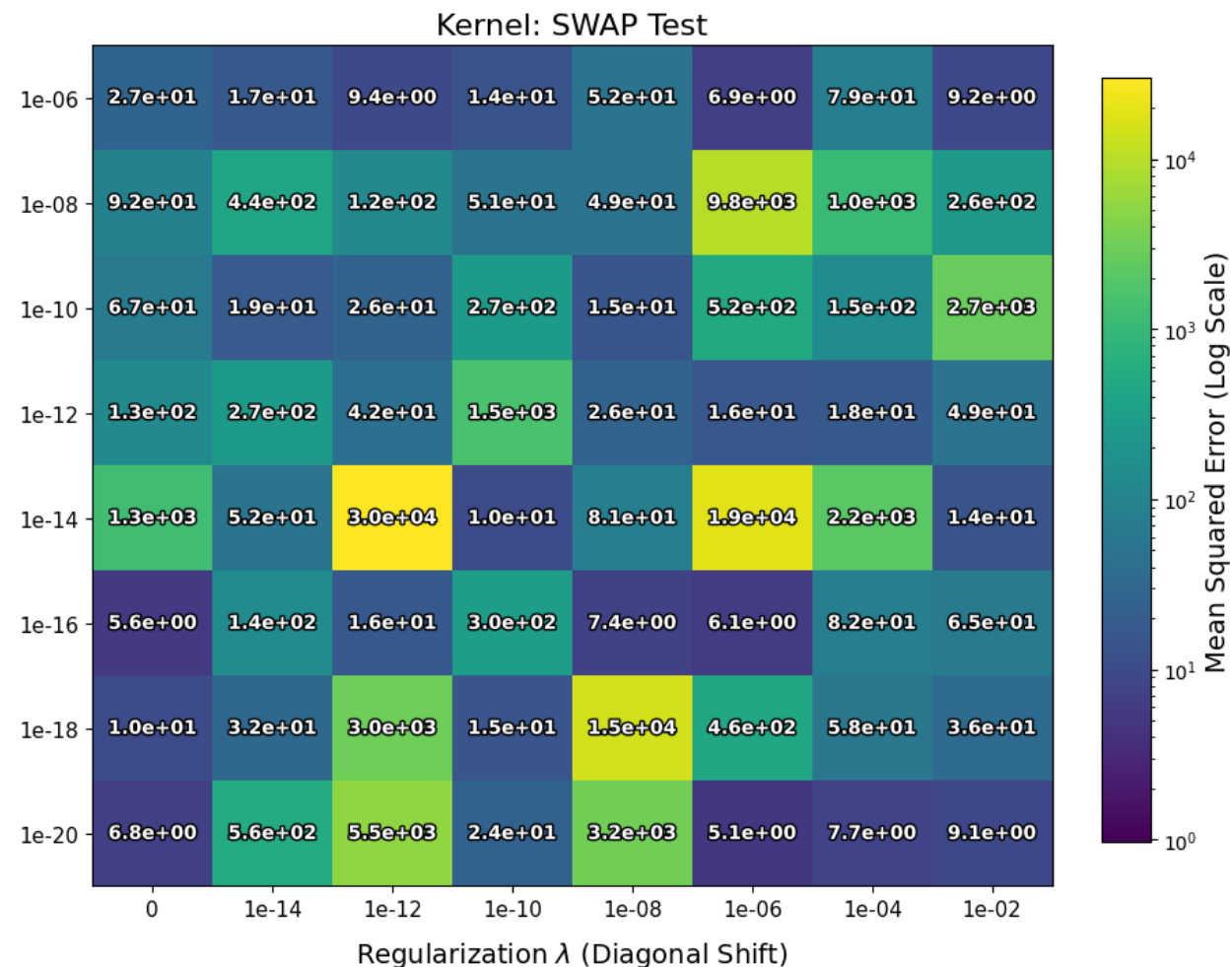
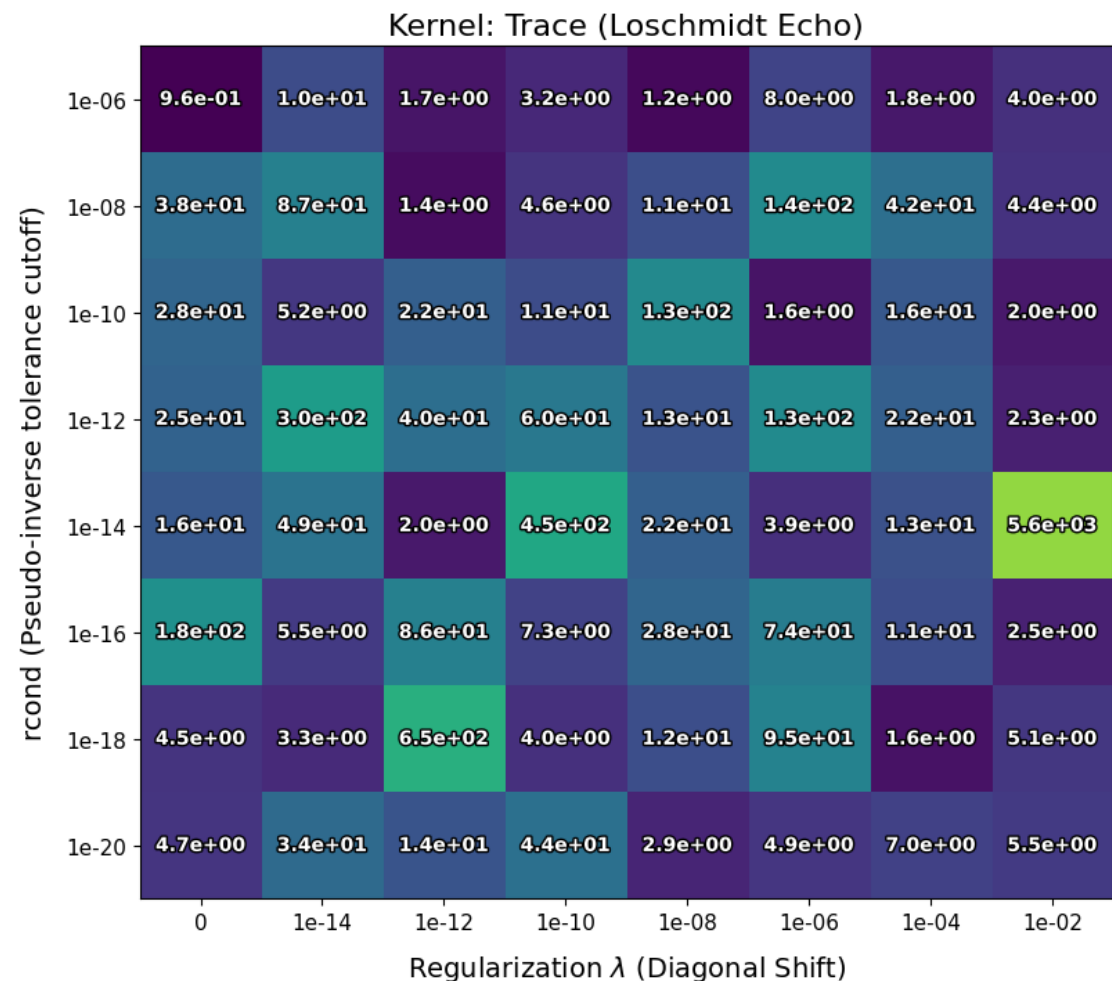
Noisy Kernel MSE, λ – rcond

Interplay of λ and rcond under Finite Shot Noise
(1000 Shots, $N_{\text{train}}=100$, 250 Realizations)



Noisy Kernel MSE, λ – rcond

Interplay of λ and rcond under Finite Shot Noise
(100 Shots, $N_{\text{train}}=100$, 250 Realizations)



Scaling (again QELM vs Kernel)

Scaling with System Size: QELM vs Quantum Kernels
(Fixed $N_{train} = 100$, 1000 Shots)

